JUNIOR SCHOOL LEARNING HUB

GRADE 8 MATHEMATICS — SCHEME OF WORK — TERM 1 2027

School: _____ Teacher: _____ TSC No.: _____

CBC planning format | Lesson outcomes are organised into Knowledge, Skills and Attitudes. Learning experiences retain the activities in the supplied Grade 8 scheme.

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
2	1	Numbers	Integers - Identification of integers Integers - Representation of integers on number line	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define integers and distinguish them from non-integers **Skills:** Identify positive integers, negative integers and zero in different situations **Attitudes:** Appreciate the use of integers in daily life situations	**In groups, learners are guided to:** • Discuss and find readings of thermometers showing positive and negative values • Classify numbers as integers or non-integers • Use real-life situations like floors above and below ground to represent integers	How do we identify integers in real life situations?	- Master Mathematics Grade 8, pg. 1 - Thermometers - Number cards - Charts with integers - Master Mathematics Grade 8, pg. 2 - Manila paper - Rulers - Markers - Number lines	- Observation - Oral questions - Written exercises	
2	2	Numbers	Integers - Addition of integers on number line Integers - Subtraction of integers on number line	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the rule for adding integers on a number line **Skills:** Carry out addition of integers on a number line correctly **Attitudes:** Value the importance of addition of integers in real life	**In groups, learners are guided to:** • Use number cards and containers for selection • Draw number lines on the ground • Jump to the right to add positive numbers • Mark and record positions after jumping	How do we carry out addition of integers?	- Master Mathematics Grade 8, pg. 3 - Number cards - Ground markings - Chalk - Counters - Master Mathematics Grade 8, pg. 4 - Number lines - Markers - Playground space	- Observation - Practical activities - Oral questions	
2	3	Numbers	Integers - Combined operations on number line Integers - Application of integers using IT resources	**At the end of the lesson, the learner should be able to:** **Knowledge:** Describe the order of combined operations on integers **Skills:** Perform combined addition and subtraction of integers on number line **Attitudes:** Show confidence in solving problems involving integers	**In groups, learners are guided to:** • Practice mixed operations using number lines • Solve problems involving temperature changes • Work out problems involving floors in buildings	How do we perform combined operations of integers?	- Master Mathematics Grade 8, pg. 5 - Number lines - Temperature gauges - Real-life problem cards - Master Mathematics Grade 8, pg. 6 - Digital devices - Internet access - Integer games/apps	- Written exercises - Problem-solving tasks - Observation	
2	4	Numbers	Fractions - Order of operations in fractions	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify the correct order of operations in fractions (BODMAS) **Skills:** Carry out combined operations on fractions accurately **Attitudes:** Appreciate the importance of following correct order	**In groups, learners are guided to:** • Discuss and use correct order of operations in fractions • Work through operations in brackets first, then multiplication/division, then addition/subtraction • Copy operations on number cards and solve	How do we use fractions in real life situations?	- Master Mathematics Grade 8, pg. 8 - Fraction cards - Calculators - Charts showing BODMAS	- Written tests - Class activities - Oral questions	
2	5	Numbers	Fractions - Operations on fractions from shopping activities Fractions - Word problems involving fractions	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain how fractions are used in shopping and trading **Skills:** Work out fraction operations from shopping activities **Attitudes:** Show responsibility in applying fractions to real situations	**In groups, learners are guided to:** • Discuss and carry out operations on fractions from shopping and other real-life cases • Role-play shopping scenarios • Solve problems involving sharing and distribution	Where do we apply combined operations on fractions?	- Master Mathematics Grade 8, pg. 9 - Shopping lists - Price tags - Play money - Fraction pieces - Master Mathematics Grade 8, pg. 10 - Word problem cards - Fraction charts - Measuring tools	- Problem-solving - Practical activities - Written assignments	
3	1	Numbers	Fractions - Games and IT activities on fractions	**At the end of the lesson, the learner should be able to:** **Knowledge:** Describe different games involving fractions **Skills:** Use IT devices for learning operations on fractions and play games **Attitudes:** Enjoy learning about fractions	**In groups, learners are guided to:** • Play games of operations on fractions using IT devices or other resources • Engage in drag and drop activities online • Use fraction apps for practice	How can we make learning fractions more interesting?	- Master Mathematics Grade 8, pg. 11 - Tablets/computers - Internet access - Fraction games	- Observation - Game performance - Digital assessment	
3	2	Numbers	Fractions - Mixed practice on combined operations	**At the end of the lesson, the learner should be able to:** **Knowledge:** Recall the order of operations in fractions **Skills:** Solve complex combined fraction operations proficiently **Attitudes:** Show confidence in working with fractions	**In groups, learners are guided to:** • Practice solving mixed fraction problems • Work in groups on challenging fraction tasks • Present solutions to the class	What strategies help us solve complex fraction problems?	- Master Mathematics Grade 8, pg. 12 - Exercise books - Fraction worksheets - Group work materials	- Written tests - Group presentations - Peer assessment	
3	3	Numbers	Fractions - Mixed practice on combined operations	**At the end of the lesson, the learner should be able to:** **Knowledge:** Recall the order of operations in fractions **Skills:** Solve complex combined fraction operations proficiently **Attitudes:** Show confidence in working with fractions	**In groups, learners are guided to:** • Practice solving mixed fraction problems • Work in groups on challenging fraction tasks • Present solutions to the class	What strategies help us solve complex fraction problems?	- Master Mathematics Grade 8, pg. 12 - Exercise books - Fraction worksheets - Group work materials	- Written tests - Group presentations - Peer assessment	
3	4	Numbers	Fractions - Application and reflection	**At the end of the lesson, the learner should be able to:** **Knowledge:** Discuss various applications of fractions in daily life **Skills:** Demonstrate mastery of fraction operations **Attitudes:** Promote use of fractions in real life situations	**In groups, learners are guided to:** • Discuss with peers, parents or guardians about areas where fractions are applied • Share real-life experiences involving fractions • Compile a portfolio of fraction work	How have fractions helped us in our daily lives?	- Master Mathematics Grade 8, pg. 13 - Portfolio materials - Reflection journals	- Portfolio assessment - Oral presentations - Self-assessment	
3	5	Numbers	Decimals - Conversion of fractions to decimals	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the relationship between fractions and decimals **Skills:** Convert fractions to decimals using different methods **Attitudes:** Appreciate the connection between fractions and decimals	**In groups, learners are guided to:** • Practice converting fractions to decimals using equivalent fractions with denominators as powers of 10 • Use long division method to convert fractions • Complete conversion tables	How do we convert fractions to decimals?	- Master Mathematics Grade 8, pg. 13 - Conversion charts - Calculators - Place value charts	- Written exercises - Oral questions - Class activities	
4	1	Numbers	Decimals - Identifying and converting recurring decimals	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define recurring and non-recurring decimals **Skills:** Identify recurring decimals and convert them to fractions correctly **Attitudes:** Show interest in working with recurring decimals	**In groups, learners are guided to:** • Discuss and classify non-recurring and recurring decimals • Indicate recurring digits using dot notation • Practice converting recurring decimals to fractions using algebraic method	How do we identify and work with recurring decimals?	- Master Mathematics Grade 8, pg. 15 - Decimal cards - Number cards - Calculators	- Written tests - Practical exercises - Observation	
4	2	Numbers	Decimals - Rounding off decimals to decimal places	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the rules for rounding off decimals **Skills:** Round off decimal numbers to required decimal places accurately **Attitudes:** Value accuracy in rounding decimals	**In groups, learners are guided to:** • Discuss and round off decimal numbers to required decimal places • Practice rounding to 1, 2, 3 decimal places • Use place value charts to understand rounding	How do we round off decimals correctly?	- Master Mathematics Grade 8, pg. 19 - Place value charts - Decimal number cards - Rounding worksheets	- Written assignments - Oral questions - Class tests	

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4	3	Numbers	Decimals - Expressing numbers in significant figures	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the meaning of significant figures **Skills:** Write decimal and whole numbers to given significant figures **Attitudes:** Show precision in expressing numbers	**In groups, learners are guided to:** • Write decimal and whole numbers to given significant figures • Discuss rules for identifying significant figures • Practice expressing numbers to different significant figures	What are significant figures and why are they important?	- Master Mathematics Grade 8, pg. 21 - Number charts - Worksheets - Scientific calculators	- Written tests - Practical exercises - Observation	
4	4	Numbers	Decimals - Expressing numbers in standard form	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define standard form notation A × 10■ **Skills:** Write numbers in standard form correctly and convert them back **Attitudes:** Appreciate the use of standard form for very large and small numbers	**In groups, learners are guided to:** • Write numbers in standard form on learning materials such as cards or charts • Practice expressing very large and very small numbers • Understand the power of 10 notation	How do we express numbers in standard form?	- Master Mathematics Grade 8, pg. 23 - Standard form cards - Calculators - Charts	- Written exercises - Oral questions - Class activities	
4	5	Numbers	Decimals - Combined operations on decimals	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify the correct order of operations for decimals **Skills:** Work out combined operations on decimals systematically **Attitudes:** Show confidence in solving decimal problems	**In groups, learners are guided to:** • Work out combined operations on decimals in the correct order • Practice problems involving brackets, multiplication, division, addition and subtraction • Solve complex decimal calculations	How do we perform combined operations on decimals?	- Master Mathematics Grade 8, pg. 24 - Operation cards - Calculators - Worksheets	- Written tests - Problem-solving - Observation	
5	1	Numbers	Decimals - Application of decimals to real life	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify situations where decimals are used in daily life **Skills:** Apply decimals to solve practical problems **Attitudes:** Promote use of decimals in daily activities	**In groups, learners are guided to:** • Discuss and apply decimals to real life cases • Solve problems involving money, measurements, temperature • Work with real-life scenarios	Where do we use decimals in our daily lives?	- Master Mathematics Grade 8, pg. 26 - Real-life problem cards - Measuring instruments - Price lists	- Practical tasks - Written assignments - Oral presentations	
5	2	Numbers	Decimals - Application of decimals to real life	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify situations where decimals are used in daily life **Skills:** Apply decimals to solve practical problems **Attitudes:** Promote use of decimals in daily activities	**In groups, learners are guided to:** • Discuss and apply decimals to real life cases • Solve problems involving money, measurements, temperature • Work with real-life scenarios	Where do we use decimals in our daily lives?	- Master Mathematics Grade 8, pg. 26 - Real-life problem cards - Measuring instruments - Price lists	- Practical tasks - Written assignments - Oral presentations	
5	3	Numbers	Decimals - Games and digital activities	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain how digital games enhance learning of decimals **Skills:** Use IT devices to play games involving decimals **Attitudes:** Enjoy learning about decimals through interactive activities	**In groups, learners are guided to:** • Play games of operations on decimals using IT or other materials • Use decimal apps and online games • Engage in interactive decimal activities	How can technology enhance our understanding of decimals?	- Master Mathematics Grade 8, pg. 27 - Digital devices - Decimal games/apps - Internet access	- Observation - Game performance - Participation	
5	4	Numbers	Squares and Square Roots - Reading squares from tables	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain how to read mathematical tables for squares **Skills:** Work out squares of numbers between 1.0 and 9.999 from tables **Attitudes:** Show accuracy in using mathematical tables	**In groups, learners are guided to:** • Read and write the squares of numbers from tables • Practice locating numbers in the table and reading their squares • Work through examples using Table 1.3	What are squares of numbers?	- Master Mathematics Grade 8, pg. 29 - Mathematical tables - Number cards - Worksheets	- Practical exercises - Written tests - Observation	
5	5	Numbers	Squares and Square Roots - Squares of large numbers	**At the end of the lesson, the learner should be able to:** **Knowledge:** Describe the method for finding squares of numbers above 10 **Skills:** Work out squares of numbers above 10 using standard form and tables **Attitudes:** Demonstrate systematic approach in calculations	**In groups, learners are guided to:** • Practice finding squares of numbers above 10 using standard form method • Convert numbers to standard form A × 10■ • Calculate squares and express in ordinary form	How do we find squares of numbers greater than 10?	- Master Mathematics Grade 8, pg. 33 - Mathematical tables - Standard form charts - Calculators	- Written exercises - Class activities - Oral questions	
6	1	Numbers	Squares and Square Roots - Squares of numbers less than 1	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the process for squaring decimal numbers less than 1 **Skills:** Find squares of decimal numbers less than 1 using tables **Attitudes:** Show precision in working with small numbers	**In groups, learners are guided to:** • Practice finding squares of numbers less than 1 • Use standard form with negative powers of 10 • Apply systematic method for calculations	How do we find squares of numbers less than 1?	- Master Mathematics Grade 8, pg. 35 - Mathematical tables - Decimal cards - Worksheets	- Written tests - Practical exercises - Problem-solving	
6	2	Numbers	Squares and Square Roots - Reading square roots from tables	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain how to read square root tables **Skills:** Work out square roots of numbers from 1 to 99.99 using tables **Attitudes:** Appreciate the relationship between squares and square roots	**In groups, learners are guided to:** • Read and write the square roots of numbers from tables • Practice using Table 1.4 for square roots • Add values from the ADD column correctly	Where do we apply square roots in real life?	- Master Mathematics Grade 8, pg. 37 - Mathematical tables - Square root charts - Number cards	- Written assignments - Oral questions - Class tests	
6	3	Numbers	Squares and Square Roots - Reading square roots from tables	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain how to read square root tables **Skills:** Work out square roots of numbers from 1 to 99.99 using tables **Attitudes:** Appreciate the relationship between squares and square roots	**In groups, learners are guided to:** • Read and write the square roots of numbers from tables • Practice using Table 1.4 for square roots • Add values from the ADD column correctly	Where do we apply square roots in real life?	- Master Mathematics Grade 8, pg. 37 - Mathematical tables - Square root charts - Number cards	- Written assignments - Oral questions - Class tests	
6	4	Numbers	Squares and Square Roots - Square roots of large numbers	**At the end of the lesson, the learner should be able to:** **Knowledge:** Describe the method for finding square roots of numbers 100 and above **Skills:** Find square roots of numbers 100 and above using tables **Attitudes:** Show systematic approach in calculations	**In groups, learners are guided to:** • Practice finding square roots of numbers above 100 • Use standard form method • Work with both Table 1.4 and Table 1.5 appropriately	How do we find square roots of numbers above 100?	- Master Mathematics Grade 8, pg. 39 - Mathematical tables (Tables 1.4 & 1.5) - Worksheets - Calculators	- Written exercises - Practical work - Observation	
6	5	Numbers	Squares and Square Roots - Using calculators for squares and square roots	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify the square and square root functions on a calculator **Skills:** Work out squares and square roots using a calculator correctly **Attitudes:** Appreciate the efficiency of using calculators	**In groups, learners are guided to:** • Practice working out squares and square roots using a calculator • Compare calculator results with table results • Use IT devices or other materials to play square and square root games	How do calculators help us find squares and square roots?	- Master Mathematics Grade 8, pg. 42 - Scientific calculators - Digital devices - Comparison worksheets	- Practical exercises - Observation - Written tests	

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7	1	Numbers	Rates, Ratio, Proportions and Percentages - Identifying rates	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define rate as a quotient relationship between two quantities **Skills:** Identify rates in different real-life situations **Attitudes:** Appreciate the use of rates in daily life	**In groups, learners are guided to:** • Time while doing different activities such as calling using different mobile service providers • Role play activities and note time taken • Record and compare rates	How do we use rates in real life situations?	- Master Mathematics Grade 8, pg. 44 - Stopwatches - Rate cards - Mobile phones (for demonstration)	- Observation - Oral questions - Practical activities	
7	2	Numbers	Rates, Ratio, Proportions and Percentages - Working out rates	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the method for calculating rates **Skills:** Calculate rates from given information accurately **Attitudes:** Show precision in rate calculations	**In groups, learners are guided to:** • Carry out activities to determine rates • Calculate rates per unit time or quantity • Solve rate problems from real-life contexts	How do we calculate rates from given information?	- Master Mathematics Grade 8, pg. 46 - Timers - Measuring tools - Rate worksheets	- Written tests - Problem-solving - Class activities	
7	3	Numbers	Rates, Ratio, Proportions and Percentages - Expressing fractions as ratios	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain how to convert fractions to ratios **Skills:** Express fractions as ratios in simplest form **Attitudes:** Value precision in ratio work	**In groups, learners are guided to:** • Use cut outs from whole objects to relate fractions to ratios • Practice writing fractions as numerator : denominator • Simplify ratios to lowest terms	How do we express fractions as ratios?	- Master Mathematics Grade 8, pg. 48 - Cut-out materials - Ratio cards - Counters	- Written exercises - Practical work - Oral questions	
7	4	Numbers	Rates, Ratio, Proportions and Percentages - Comparing ratios	**At the end of the lesson, the learner should be able to:** **Knowledge:** Describe methods for comparing two or more ratios **Skills:** Compare ratios using percentage method and LCM method **Attitudes:** Show systematic approach in comparing ratios	**In groups, learners are guided to:** • Discuss and compare ratios from cut outs • Use LCM method to compare ratios • Express ratios as percentages for easy comparison	How do we compare two or more ratios?	- Master Mathematics Grade 8, pg. 50 - Comparison charts - Ratio cards - Calculators	- Written tests - Class activities - Problem-solving	
7	5	Numbers	Rates, Ratio, Proportions and Percentages - Comparing ratios	**At the end of the lesson, the learner should be able to:** **Knowledge:** Describe methods for comparing two or more ratios **Skills:** Compare ratios using percentage method and LCM method **Attitudes:** Show systematic approach in comparing ratios	**In groups, learners are guided to:** • Discuss and compare ratios from cut outs • Use LCM method to compare ratios • Express ratios as percentages for easy comparison	How do we compare two or more ratios?	- Master Mathematics Grade 8, pg. 50 - Comparison charts - Ratio cards - Calculators	- Written tests - Class activities - Problem-solving	
8	1	Numbers	Rates, Ratio, Proportions and Percentages - Division of quantities in ratios	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the process of dividing quantities in given ratios **Skills:** Divide quantities in given ratios systematically **Attitudes:** Show fairness in sharing quantities	**In groups, learners are guided to:** • Discuss and share quantities of concrete objects in different ratios • Use counters or bottle tops to practice sharing • Solve sharing problems	How do we divide quantities using ratios?	- Master Mathematics Grade 8, pg. 51 - Counters - Bottle tops - Sharing materials	- Practical exercises - Written assignments - Observation	
8	2	Numbers	Rates, Ratio, Proportions and Percentages - Working out ratios	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify the method for finding ratios from given quantities **Skills:** Work out ratios in different situations **Attitudes:** Appreciate applications of ratios in daily life	**In groups, learners are guided to:** • Calculate ratios from given quantities • Find missing values in ratio problems • Apply ratios to real situations	How do we determine ratios from given information?	- Master Mathematics Grade 8, pg. 53 - Data cards - Real-life examples - Worksheets	- Written tests - Problem-solving - Oral questions	
8	3	Numbers	Rates, Ratio, Proportions and Percentages - Increase and decrease using ratios	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain how ratios show increase or decrease in quantities **Skills:** Work out increase and decrease of quantities using ratios **Attitudes:** Apply ratio changes to real situations	**In groups, learners are guided to:** • Discuss and determine increase and decrease using ratios • Use the format new : old to express changes • Solve problems involving ratio changes	How do ratios represent increase or decrease?	- Master Mathematics Grade 8, pg. 55 - Change scenario cards - Calculators - Worksheets	- Written exercises - Class activities - Problem-solving	
8	4	Numbers	Rates, Ratio, Proportions and Percentages - Percentage increase	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define percentage increase **Skills:** Calculate percentage increase accurately using the formula **Attitudes:** Show precision in percentage calculations	**In groups, learners are guided to:** • Discuss and determine percentage increase of different quantities • Use the formula: percentage change = (change/original) × 100% • Solve real-life percentage problems	How do we calculate percentage increase?	- Master Mathematics Grade 8, pg. 57 - Percentage charts - Calculators - Problem cards	- Written tests - Practical exercises - Oral questions	
8	5	Numbers	Rates, Ratio, Proportions and Percentages - Percentage decrease	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define percentage decrease **Skills:** Calculate percentage decrease correctly **Attitudes:** Apply percentage decrease to real situations responsibly	**In groups, learners are guided to:** • Work through percentage decrease problems • Calculate new values after percentage decrease • Solve problems involving discounts and reductions	How do we calculate percentage decrease?	- Master Mathematics Grade 8, pg. 58 - Discount cards - Price lists - Calculators	- Written assignments - Problem-solving - Class tests	
9	1	Numbers	Rates, Ratio, Proportions and Percentages - Identifying direct proportions	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define direct proportion **Skills:** Identify direct proportions in real life situations **Attitudes:** Appreciate proportional relationships in daily activities	**In groups, learners are guided to:** • Use IT devices or other materials to explore proportions • Role play shopping activities to show direct relationships • Identify situations where increase in one leads to increase in other	What is direct proportion?	- Master Mathematics Grade 8, pg. 59 - Proportion charts - Real-life examples - Digital devices	- Observation - Oral questions - Practical activities	
9	2	Numbers	Rates, Ratio, Proportions and Percentages - Working out direct proportions	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the unitary method for solving direct proportion **Skills:** Work out direct proportions systematically **Attitudes:** Show accuracy in direct proportion calculations	**In groups, learners are guided to:** • Complete tables showing direct proportional relationships • Calculate missing values in direct proportion • Apply direct proportion to solve problems	How do we solve direct proportion problems?	- Master Mathematics Grade 8, pg. 60 - Proportion tables - Worksheets - Calculators	- Written tests - Problem-solving - Class activities	
9	3	Numbers	Rates, Ratio, Proportions and Percentages - Working out direct proportions	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the unitary method for solving direct proportion **Skills:** Work out direct proportions systematically **Attitudes:** Show accuracy in direct proportion calculations	**In groups, learners are guided to:** • Complete tables showing direct proportional relationships • Calculate missing values in direct proportion • Apply direct proportion to solve problems	How do we solve direct proportion problems?	- Master Mathematics Grade 8, pg. 60 - Proportion tables - Worksheets - Calculators	- Written tests - Problem-solving - Class activities	
9	4	Numbers	Rates, Ratio, Proportions and Percentages - Identifying indirect proportions	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define indirect proportion **Skills:** Identify indirect proportions in different situations **Attitudes:** Appreciate the difference between direct and indirect proportion	**In groups, learners are guided to:** • Use hourglass to show and determine indirect relationships • Identify situations where increase in one leads to decrease in other • Practice with filling containers	What is indirect proportion?	- Master Mathematics Grade 8, pg. 62 - Hourglass - Containers - Bottle tops	- Observation - Practical work - Oral questions	
9	5	Numbers	Rates, Ratio, Proportions and Percentages - Working out indirect proportions	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the method for solving indirect proportion **Skills:** Work out indirect proportions systematically **Attitudes:** Show understanding of inverse relationships	**In groups, learners are guided to:** • Complete tables showing indirect proportional relationships • Calculate values where ratios are inverted • Solve time-speed-distance problems	How do we solve indirect proportion problems?	- Master Mathematics Grade 8, pg. 63 - Proportion worksheets - Calculators - Problem cards	- Written exercises - Problem-solving - Written tests	

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10	1	Numbers	Rates, Ratio, Proportions and Percentages - Application and reflection	**At the end of the lesson, the learner should be able to:** **Knowledge:** Discuss various applications of ratios and proportions **Skills:** Apply ratios and proportions in various real-life contexts **Attitudes:** Promote use of ratios and proportions in real life	**In groups, learners are guided to:** • Watch videos on ratios and proportions as used in daily activities • Discuss applications with parents or guardians • Reflect on learning and compile portfolio	How do ratios and proportions help us in daily life?	- Master Mathematics Grade 8, pg. 64 - Video resources - Digital devices - Portfolio materials	- Portfolio assessment - Presentations - Self-assessment	
10	2	Algebra	Algebraic Expressions - Factorisation of algebraic expressions	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define factorisation as the reverse of expansion **Skills:** Identify the highest common factor (HCF) in algebraic expressions **Attitudes:** Appreciate the use of factorisation in simplifying expressions	**In groups, learners are guided to:** • Make three sets of cards showing algebraic expressions and their factored forms • Match cards from different rows to form equations • Discuss and identify common factors in terms • Write HCF in front of brackets and remaining factors inside	How do we factorise algebraic expressions?	- Master Mathematics Grade 8, pg. 65 - Number cards - Algebraic expression cards - Charts	- Observation - Card matching activity - Oral questions	
10	3	Algebra	Algebraic Expressions - Identifying like and unlike terms in factorisation	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the concept of like and unlike terms **Skills:** Find common factors for different sets of terms **Attitudes:** Show systematic approach in identifying factors	**In groups, learners are guided to:** • Discuss and identify like and unlike terms • Find common factors from given sets of algebraic terms • Practice factorising expressions with numerical and variable common factors • Work in groups to factorise various expressions	What makes terms like or unlike in algebra?	- Master Mathematics Grade 8, pg. 67 - Factor cards - Worksheets - Group work materials	- Written exercises - Group presentations - Class activities	
10	4	Algebra	Algebraic Expressions - Simplification of algebraic fractions	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the process of simplifying algebraic fractions **Skills:** Simplify algebraic fractions by finding LCM of denominators **Attitudes:** Value accuracy in simplifying fractions	**In groups, learners are guided to:** • Discuss like and unlike terms in algebraic fractions • Find LCM of denominators in algebraic fractions • Combine fractions with different denominators • Practice simplifying complex algebraic fractions	How do we simplify algebraic expressions?	- Master Mathematics Grade 8, pg. 68 - Fraction charts - LCM charts - Worksheets	- Written tests - Practical exercises - Problem-solving	
10	5	Algebra	Algebraic Expressions - Simplification of algebraic fractions	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the process of simplifying algebraic fractions **Skills:** Simplify algebraic fractions by finding LCM of denominators **Attitudes:** Value accuracy in simplifying fractions	**In groups, learners are guided to:** • Discuss like and unlike terms in algebraic fractions • Find LCM of denominators in algebraic fractions • Combine fractions with different denominators • Practice simplifying complex algebraic fractions	How do we simplify algebraic expressions?	- Master Mathematics Grade 8, pg. 68 - Fraction charts - LCM charts - Worksheets	- Written tests - Practical exercises - Problem-solving	
11	1	Algebra	Algebraic Expressions - Advanced simplification practice	**At the end of the lesson, the learner should be able to:** **Knowledge:** Describe steps for simplifying complex algebraic fractions **Skills:** Simplify algebraic fractions involving multiple operations **Attitudes:** Show confidence in working with algebraic fractions	**In groups, learners are guided to:** • Practice writing fractions as single fractions • Simplify fractions with algebraic denominators • Solve problems involving algebraic fractions • Work through real-life applications	What strategies help us simplify complex algebraic fractions?	- Master Mathematics Grade 8, pg. 69 - Practice worksheets - Real-life problem cards - Calculators	- Written assignments - Class tests - Oral questions	
11	2	Algebra	Algebraic Expressions - Using IT devices and application	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify IT resources for learning algebra **Skills:** Use IT devices to work out algebra exercises and drag-drop activities **Attitudes:** Enjoy using algebraic expressions in real life situations	**In groups, learners are guided to:** • Use IT devices to work out exercises and activities in algebra • Engage in drag and drop activities of grouping similar terms • Play online games simplifying algebraic expressions • Discuss applications with peers and parents	How can technology enhance our understanding of algebra?	- Master Mathematics Grade 8, pg. 71 - Digital devices - Internet access - Algebra apps/software	- Observation - Digital assessment - Participation	
11	3	Algebra	Linear Equations - Forming linear equations in two unknowns	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define linear equations in two unknowns **Skills:** Form linear equations from real-life situations using two variables **Attitudes:** Show interest in forming equations from word problems	**In groups, learners are guided to:** • Put masses on beam balance and add marbles to balance • Give letters to represent unknowns • Role play shopping activities to form equations • Write equations from balancing scenarios	How do we solve linear equations in two unknowns?	- Master Mathematics Grade 8, pg. 72 - Beam balance - Masses (500g) - Marbles - Shopping scenario cards	- Observation - Practical activities - Oral questions	
11	4	Algebra	Linear Equations - More practice on forming equations	**At the end of the lesson, the learner should be able to:** **Knowledge:** Interpret word problems involving two unknowns **Skills:** Form linear equations from various real-life scenarios **Attitudes:** Appreciate the relevance of equations in daily life	**In groups, learners are guided to:** • Write equations to represent ages, costs, and quantities • Form equations from perimeter problems • Create equations from problems involving animals and farming • Practice with two-digit number problems	Where do we use linear equations in two unknowns in real life situations?	- Master Mathematics Grade 8, pg. 73 - Word problem cards - Real-life scenario cards - Worksheets	- Written exercises - Problem-solving - Class activities	
11	5	Algebra	Linear Equations - Solving by substitution method	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the substitution method for solving simultaneous equations **Skills:** Solve linear equations in two unknowns using substitution systematically **Attitudes:** Show precision in solving equations	**In groups, learners are guided to:** • Write equations from fruit vendor scenario • Name equations as (i) and (ii) • Write one variable in terms of another • Replace and simplify to find values of unknowns	How do we use substitution method to solve linear equations?	- Master Mathematics Grade 8, pg. 74 - Fruit pictures - Equation cards - Step-by-step charts	- Written tests - Practical exercises - Oral questions	
12	1	Algebra	Linear Equations - Advanced practice on substitution method	**At the end of the lesson, the learner should be able to:** **Knowledge:** Describe the complete process of substitution method **Skills:** Solve complex simultaneous equations by substitution **Attitudes:** Demonstrate mastery of substitution technique	**In groups, learners are guided to:** • Practice solving equations with fractions using substitution • Work through problems involving costs and quantities • Solve problems about carpentry and furniture making • Apply substitution to number problems	What are the key steps in substitution method?	- Master Mathematics Grade 8, pg. 75 - Practice worksheets - Real-life problem cards - Calculators	- Written assignments - Problem-solving - Class tests	
12	2	Algebra	Linear Equations - Advanced practice on substitution method	**At the end of the lesson, the learner should be able to:** **Knowledge:** Describe the complete process of substitution method **Skills:** Solve complex simultaneous equations by substitution **Attitudes:** Demonstrate mastery of substitution technique	**In groups, learners are guided to:** • Practice solving equations with fractions using substitution • Work through problems involving costs and quantities • Solve problems about carpentry and furniture making • Apply substitution to number problems	What are the key steps in substitution method?	- Master Mathematics Grade 8, pg. 75 - Practice worksheets - Real-life problem cards - Calculators	- Written assignments - Problem-solving - Class tests	
12	3	Algebra	Linear Equations - Solving by elimination method	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the elimination method for solving simultaneous equations **Skills:** Solve linear equations using elimination method systematically **Attitudes:** Appreciate the efficiency of elimination method	**In groups, learners are guided to:** • Form equations from shopping scenarios (plates and cups) • Multiply equations to make coefficients equal • Subtract corresponding parts to eliminate one variable • Solve for remaining variable and substitute back	How do we solve equations using elimination method?	- Master Mathematics Grade 8, pg. 76 - Shopping scenario cards - Elimination charts - Step-by-step guides	- Written exercises - Practical work - Oral questions	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
12	4	Algebra	Linear Equations - More practice on elimination method	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify when to use elimination method **Skills:** Solve various simultaneous equations by elimination efficiently **Attitudes:** Show confidence in choosing appropriate methods	**In groups, learners are guided to:** • Practice solving equations involving bread and tea leaves • Work through problems with different coefficients • Solve problems about costs of items • Compare elimination and substitution methods	When is elimination method more suitable than substitution?	- Master Mathematics Grade 8, pg. 78 - Comparison charts - Practice worksheets - Method selection guides	- Written tests - Class activities - Problem-solving	
12	5	Algebra	Linear Equations - Application in real-life situations	**At the end of the lesson, the learner should be able to:** **Knowledge:** Discuss various applications of linear equations in daily life **Skills:** Apply linear equations to solve real-life problems involving rectangles, costs, and quantities **Attitudes:** Recognize use of linear equations in real life	**In groups, learners are guided to:** • Find sum and difference of two numbers using equations • Solve problems about rectangular flower beds • Work out problems involving hiring labourers • Apply equations to school fees and shopping scenarios • Watch videos on linear equations applications	How do linear equations help us solve real-life problems?	- Master Mathematics Grade 8, pg. 79 - Video resources - Real-life scenario cards - Digital devices - Application worksheets	- Portfolio assessment - Presentations - Written assignments - Self-assessment	

JUNIOR SCHOOL LEARNING HUB

GRADE 8 MATHEMATICS — SCHEME OF WORK — TERM 2 2027

School: _____ Teacher: _____ TSC No.: _____

CBC planning format | Lesson outcomes are organised into Knowledge, Skills and Attitudes. Learning experiences retain the activities in the supplied Grade 8 scheme.

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
2	1	Measurements	Circles - Circumference of a circle	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define circumference as the distance around a circle **Skills:** Calculate the circumference using the formula C=πD or C=2πr **Attitudes:** Appreciate the relationship between diameter and circumference	**In groups, learners are guided to:** • Take a string and two sticks to draw circles on the ground • Measure the distance between fixed points • Use string and ruler to measure total length of line drawn • Compare diameter measurement with circumference	How do we determine the circumference of a circle?	- Master Mathematics Grade 8, pg. 81 - Strings - Sticks - Rulers - Circular objects	- Practical activities - Oral questions - Written exercises	
2	2	Measurements	Circles - Finding circumference of circular objects	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify circular objects in the environment **Skills:** Work out the circumference of different circular objects accurately **Attitudes:** Show interest in measuring circular objects	**In groups, learners are guided to:** • Discuss and find circumference of different circular objects in the environment • Complete tables to find missing measurements (radius, diameter, circumference) • Calculate circumference of bicycle wheels and clock hands • Solve real-life problems involving wheels and revolutions	Where do we find circles in our environment?	- Master Mathematics Grade 8, pg. 82 - Bicycle wheels - Clock models - Measuring tape - Circular objects	- Written tests - Practical work - Problem-solving	
2	3	Measurements	Circles - Length of an arc	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define an arc as a portion of circumference **Skills:** Calculate arc length using the formula Arc length = (θ/360) × 2πr **Attitudes:** Value the importance of arc calculations in real life	**In groups, learners are guided to:** • Make dummy clock using available resources • Trace path of minute hand in one revolution • Measure angles at centre and calculate arc lengths • Use cut outs to relate arcs to sectors	How do we calculate the length of an arc?	- Master Mathematics Grade 8, pg. 84 - Cartons for clock - Protractors - Strings - Rulers	- Practical exercises - Written assignments - Oral questions	
2	4	Measurements	Circles - Length of an arc	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define an arc as a portion of circumference **Skills:** Calculate arc length using the formula Arc length = (θ/360) × 2πr **Attitudes:** Value the importance of arc calculations in real life	**In groups, learners are guided to:** • Make dummy clock using available resources • Trace path of minute hand in one revolution • Measure angles at centre and calculate arc lengths • Use cut outs to relate arcs to sectors	How do we calculate the length of an arc?	- Master Mathematics Grade 8, pg. 84 - Cartons for clock - Protractors - Strings - Rulers	- Practical exercises - Written assignments - Oral questions	
2	5	Measurements	Circles - Perimeter of a sector	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain what a sector is and identify minor and major sectors **Skills:** Calculate perimeter of a sector using the formula: Perimeter = (θ/360 × 2πr) + 2r **Attitudes:** Show systematic approach in calculating sector perimeters	**In groups, learners are guided to:** • Draw circles and mark points to form sectors • Use string and ruler to determine arc length and add radii • Measure angles at centre • Calculate perimeter using formula and compare with measured values	How do we calculate the perimeter of a sector?	- Master Mathematics Grade 8, pg. 86 - Drawing instruments - Strings - Rulers - Protractors	- Written tests - Class activities - Problem-solving	
3	1	Measurements	Circles - Application and use of IT resources	**At the end of the lesson, the learner should be able to:** **Knowledge:** Discuss various applications of circles in real life **Skills:** Use IT or other resources to explore use of sectors and arcs **Attitudes:** Promote use of circles in real life situations	**In groups, learners are guided to:** • Solve problems involving merry-go-rounds, shot put areas • Calculate perimeters of semicircular objects • Use IT devices to explore circle applications • Work on complex problems involving multiple circles	How do we use circles in real life situations?	- Master Mathematics Grade 8, pg. 87 - Digital devices - Internet access - Real-life scenario cards	- Portfolio assessment - Presentations - Written assignments	
3	2	Measurements	Area - Area of a circle	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain how the formula for area of circle is derived **Skills:** Calculate area of a circle using the formula A = πr^2 **Attitudes:** Appreciate the importance of knowing circle areas	**In groups, learners are guided to:** • Draw and cut circles into equal sections • Arrange sections to form rectangle-like shape • Relate sides of rectangle to radius of circle • Work out area of rectangle formed	How do we calculate the area of a circle?	- Master Mathematics Grade 8, pg. 88 - Plain paper - Scissors - Rulers - Circular cut-outs	- Practical work - Written exercises - Oral questions	
3	3	Measurements	Area - Calculating areas of circles with different radii	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the formula for area of a circle **Skills:** Calculate areas of circles given radius or diameter **Attitudes:** Show accuracy in area calculations	**In groups, learners are guided to:** • Calculate areas of circles with various radii • Find radius when area is given • Solve problems involving circular mats and grazing fields • Work out problems involving wire reshaping	What is the relationship between radius and area?	- Master Mathematics Grade 8, pg. 89 - Calculators - Worksheets - Problem cards	- Written tests - Problem-solving - Class activities	
3	4	Measurements	Area - Area of a sector of a circle	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define a sector as a fraction of a circle **Skills:** Calculate area of a sector using the formula: Area = (θ/360) × πr^2 **Attitudes:** Value precision in sector calculations	**In groups, learners are guided to:** • Draw circles and fold into equal parts • Calculate area using angle and radius • Use formula to find sector areas • Compare calculated areas with measured areas	How do we find the area of a sector?	- Master Mathematics Grade 8, pg. 91 - Drawing instruments - Protractors - Calculators - Paper for folding	- Written exercises - Practical activities - Oral questions	
3	5	Measurements	Area - Area of a sector of a circle	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define a sector as a fraction of a circle **Skills:** Calculate area of a sector using the formula: Area = (θ/360) × πr^2 **Attitudes:** Value precision in sector calculations	**In groups, learners are guided to:** • Draw circles and fold into equal parts • Calculate area using angle and radius • Use formula to find sector areas • Compare calculated areas with measured areas	How do we find the area of a sector?	- Master Mathematics Grade 8, pg. 91 - Drawing instruments - Protractors - Calculators - Paper for folding	- Written exercises - Practical activities - Oral questions	
4	1	Measurements	Area - Surface area of cubes	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain that a cube has 6 equal square faces **Skills:** Calculate total surface area using formula: TSA = 6 × length × length **Attitudes:** Show understanding of closed and open cubes	**In groups, learners are guided to:** • Study cubes and count number of faces • Measure sides of each face • Calculate area of each face • Derive formula for surface area of closed and open cubes	How do we calculate surface area of cubes?	- Master Mathematics Grade 8, pg. 92 - Cube models - Rulers - Measuring tape - Worksheets	- Written tests - Practical work - Problem-solving	
4	2	Measurements	Area - Surface area of cuboids	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify that cuboids have three pairs of equal rectangular faces **Skills:** Calculate surface area of cuboids systematically **Attitudes:** Appreciate applications of cuboid surface areas	**In groups, learners are guided to:** • Pick textbooks and measure length, width, height • Calculate area of each surface • Use models to understand pairs of equal sides • Derive formula for surface area	How is surface area of cuboid different from cube?	- Master Mathematics Grade 8, pg. 94 - Cuboid objects - Rulers - Cartons - Measuring instruments	- Written assignments - Class activities - Oral questions	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
4	3	Measurements	Area - Surface area of cylinders	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain that a cylinder opens to form two circles and a rectangle **Skills:** Calculate curved surface area using formula: CSA = 2πrh **Attitudes:** Show systematic approach in cylinder calculations	**In groups, learners are guided to:** • Select paper or plastic cylinders • Cut out top and bottom circles • Slit open hollow cylindrical part • Measure opened figure and relate to circumference	How do we find surface area of cylinders?	- Master Mathematics Grade 8, pg. 97 - Cylindrical objects - Scissors - Rulers - Paper cylinders	- Practical exercises - Written tests - Problem-solving	
4	4	Measurements	Area - Closed and open cylinders	**At the end of the lesson, the learner should be able to:** **Knowledge:** Distinguish between closed, open cylinders and pipes **Skills:** Calculate total surface area including circular ends **Attitudes:** Apply formulas to solve real-life problems	**In groups, learners are guided to:** • Calculate total surface area of closed cylinders • Work out surface area of open tanks and pipes • Solve problems involving petrol tanks and water pipes • Calculate surface area of semi-cylindrical troughs	When do we use different cylinder formulas?	- Master Mathematics Grade 8, pg. 99 - Cylinder models - Calculators - Real-life scenario cards	- Written assignments - Problem-solving - Class tests	
4	5	Measurements	Area - Closed and open cylinders	**At the end of the lesson, the learner should be able to:** **Knowledge:** Distinguish between closed, open cylinders and pipes **Skills:** Calculate total surface area including circular ends **Attitudes:** Apply formulas to solve real-life problems	**In groups, learners are guided to:** • Calculate total surface area of closed cylinders • Work out surface area of open tanks and pipes • Solve problems involving petrol tanks and water pipes • Calculate surface area of semi-cylindrical troughs	When do we use different cylinder formulas?	- Master Mathematics Grade 8, pg. 99 - Cylinder models - Calculators - Real-life scenario cards	- Written assignments - Problem-solving - Class tests	
5	1	Measurements	Area - Surface area of triangular prisms	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify the faces that make up a triangular prism **Skills:** Calculate surface area as sum of individual faces **Attitudes:** Value accuracy in prism calculations	**In groups, learners are guided to:** • Study triangular prism objects • Count number of faces • Identify triangular and rectangular faces • Calculate area of each face and find total	How do we calculate surface area of triangular prisms?	- Master Mathematics Grade 8, pg. 100 - Prism models - Rulers - Measuring instruments - Worksheets	- Written tests - Practical work - Oral questions	
5	2	Measurements	Area - Applications of triangular prisms	**At the end of the lesson, the learner should be able to:** **Knowledge:** Discuss real-life objects in the shape of triangular prisms **Skills:** Calculate surface areas of dust pans, tents, and goal posts **Attitudes:** Show interest in applying prism knowledge	**In groups, learners are guided to:** • Calculate surface area of rabbit hutches • Work out surface area of tents and dust pans • Solve problems involving wedges • Calculate surface area of handball goal posts covered with nets	Where do we find triangular prisms in real life?	- Master Mathematics Grade 8, pg. 102 - Real-life problem cards - Prism models - Calculators	- Written assignments - Problem-solving - Presentations	
5	3	Measurements	Area - Area of irregular shapes using square grids	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the method for estimating area of irregular shapes **Skills:** Estimate areas by counting full and partial squares **Attitudes:** Show patience in counting and estimating	**In groups, learners are guided to:** • Select graph paper and trace leaf outlines • Count number of full squares enclosed • Count partial squares and divide by 2 • Add full squares to half of partial squares	How do we estimate areas of irregular shapes?	- Master Mathematics Grade 8, pg. 103 - Graph paper - Square grids - Leaves - Pencils	- Practical activities - Written exercises - Observation	
5	4	Measurements	Area - Estimating areas of maps and other irregular shapes	**At the end of the lesson, the learner should be able to:** **Knowledge:** Apply square grid method to various irregular shapes **Skills:** Estimate areas of maps, assembly zones, and hand traces **Attitudes:** Promote use of area estimation in real life	**In groups, learners are guided to:** • Estimate area of fire assembly zones • Work out area of constituency maps • Estimate area of Kenya map • Trace palm of hand and estimate its area	What are practical uses of estimating irregular areas?	- Master Mathematics Grade 8, pg. 105 - Graph paper - Maps - Tracing paper - Calculators	- Portfolio assessment - Practical work - Written assignments	
5	5	Measurements	Money - Interest and principal	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define interest as extra money paid on borrowed amount **Skills:** Define principal as money borrowed **Attitudes:** Appreciate understanding of financial terms	**In groups, learners are guided to:** • Discuss amount of money that can be borrowed from mobile money providers • Calculate difference between amount borrowed and paid back • Identify institutions that offer loans • Complete tables relating principal, interest and total amount	What is interest in money?	- Master Mathematics Grade 8, pg. 107 - Sample loan documents - Calculators - Financial scenario cards	- Written exercises - Oral questions - Class activities	
6	1	Measurements	Money - Calculating simple interest	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain simple interest as money charged only on principal **Skills:** Calculate simple interest using formula: S.I = P × R × T / 100 **Attitudes:** Show accuracy in simple interest calculations	**In groups, learners are guided to:** • Discuss Mr. Murithi's loan scenario • Calculate total amount paid and interest • Express interest as percentage • Practice using formula with different values	How do we calculate simple interest?	- Master Mathematics Grade 8, pg. 109 - Calculators - Formula charts - Loan scenario cards	- Written tests - Problem-solving - Class activities	
6	2	Measurements	Money - Calculating simple interest	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain simple interest as money charged only on principal **Skills:** Calculate simple interest using formula: S.I = P × R × T / 100 **Attitudes:** Show accuracy in simple interest calculations	**In groups, learners are guided to:** • Discuss Mr. Murithi's loan scenario • Calculate total amount paid and interest • Express interest as percentage • Practice using formula with different values	How do we calculate simple interest?	- Master Mathematics Grade 8, pg. 109 - Calculators - Formula charts - Loan scenario cards	- Written tests - Problem-solving - Class activities	
6	3	Measurements	Money - Applications of simple interest	**At the end of the lesson, the learner should be able to:** **Knowledge:** Discuss various situations where simple interest applies **Skills:** Calculate amount paid back including interest **Attitudes:** Apply simple interest to solve real-life problems	**In groups, learners are guided to:** • Calculate interest for businessmen borrowing from financial institutions • Work out amount in bank accounts after interest • Find rate of simple interest from given information • Calculate interest earned on deposits	Where do we use simple interest in real life?	- Master Mathematics Grade 8, pg. 110 - Calculators - Real-life problem cards - Bank documents (samples)	- Written assignments - Problem-solving - Oral presentations	
6	4	Measurements	Money - Compound interest calculation step by step	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define compound interest as interest on principal and previous interest **Skills:** Calculate compound interest year by year up to three years **Attitudes:** Value systematic approach in compound interest	**In groups, learners are guided to:** • Discuss Mrs. Rono's investment in women groups • Calculate interest for first year and add to principal • Use new total as principal for second year • Continue process up to three years	How is compound interest different from simple interest?	- Master Mathematics Grade 8, pg. 112 - Calculators - Step-by-step charts - Comparison worksheets	- Written tests - Practical exercises - Class tests	
6	5	Measurements	Money - Working out appreciation per annum	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define appreciation as gain in value of a commodity **Skills:** Calculate appreciation using compound interest method **Attitudes:** Show understanding that appreciation is calculated like compound interest	**In groups, learners are guided to:** • Discuss meaning of appreciation in relation to monetary value • List items that appreciate in value • Calculate appreciation of land value year by year • Apply appreciation formula to various scenarios	What items appreciate in value and why?	- Master Mathematics Grade 8, pg. 115 - Calculators - Appreciation scenario cards - Charts	- Written exercises - Problem-solving - Oral questions	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
7	1	Measurements	Money - Working out depreciation per annum	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define depreciation as loss in value of a commodity **Skills:** Calculate depreciation step by step up to three years **Attitudes:** Appreciate that depreciation helps in making purchasing decisions	**In groups, learners are guided to:** • Discuss items that depreciate in value • Calculate depreciation of vehicles and electronics • Work through depreciation year by year • Compare depreciation with appreciation	What is depreciation and how do we calculate it?	- Master Mathematics Grade 8, pg. 116 - Calculators - Depreciation charts - Real-life examples	- Written tests - Class activities - Problem-solving	
7	2	Measurements	Money - Working out depreciation per annum	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define depreciation as loss in value of a commodity **Skills:** Calculate depreciation step by step up to three years **Attitudes:** Appreciate that depreciation helps in making purchasing decisions	**In groups, learners are guided to:** • Discuss items that depreciate in value • Calculate depreciation of vehicles and electronics • Work through depreciation year by year • Compare depreciation with appreciation	What is depreciation and how do we calculate it?	- Master Mathematics Grade 8, pg. 116 - Calculators - Depreciation charts - Real-life examples	- Written tests - Class activities - Problem-solving	
7	3	Measurements	Money - Hire purchase	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain hire purchase as buying goods through installments **Skills:** Calculate total cost under hire purchase **Attitudes:** Show consumer awareness in comparing cash and hire purchase prices	**In groups, learners are guided to:** • Visit places offering hire purchase or do online searches • Discuss different terms of purchase • Calculate installment periods and total amounts • Compare hire purchase prices with cash prices for consumer protection	How do we pay for goods on hire purchase?	- Master Mathematics Grade 8, pg. 117 - Hire purchase documents - Price comparison charts - Calculators	- Written assignments - Research projects - Oral presentations	
7	4	Measurements	Money - Visiting financial institutions and using IT for shopping	**At the end of the lesson, the learner should be able to:** **Knowledge:** Discuss information gathered from financial institutions **Skills:** Use IT to access online shopping platforms and identify terms of sale **Attitudes:** Spend money responsibly on needs and leisure	**In groups, learners are guided to:** • Visit or invite resource persons from banks and SACCOs • Gather information about interest rates offered on deposits • Use IT to access online shopping platforms • Discuss terms of sale for consumer awareness and protection	How do we make informed financial decisions?	- Master Mathematics Grade 8, pg. 118 - Digital devices - Internet access - Financial institution brochures - Guest speakers	- Portfolio assessment - Presentations - Reflection journals - Self-assessment	
7	5	4.0: Geometry	4.1: Geometrical Constructions - Constructing parallel lines using ruler and compasses 4.1: Geometrical Constructions - Constructing parallel lines using set square and ruler	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define parallel lines **Skills:** Construct parallel lines using a ruler and pair of compasses **Attitudes:** Appreciate the importance of accurate geometric constructions	**In groups, learners are guided to:** • Discuss the concept of parallel lines in real life • Follow step-by-step construction procedure using compass arcs • Draw a line and mark a point above it • Use compass arcs to construct parallel line through the point • Compare constructed lines with classmates	How can we construct parallel lines without measuring angles?	- MASTER Mathematics Grade 8 Learner's Book pg. 119 - Ruler - Pair of compasses - Pencil - Plain paper - Set square - Drawing paper	- Observation - Practical construction tasks - Oral questions	
8	1	4.0: Geometry	4.1: Geometrical Constructions - Constructing perpendicular bisector of a line 4.1: Geometrical Constructions - Constructing perpendicular from a point to a line using compasses	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define perpendicular bisector **Skills:** Construct perpendicular bisector using ruler and compasses **Attitudes:** Value accuracy in constructions	**In groups, learners are guided to:** • Draw a line of given length • Use compass to mark arcs from both ends • Identify intersection points of arcs • Join intersection points to form perpendicular bisector • Measure and verify equal segments and right angles	Why is the perpendicular bisector important in geometry?	- MASTER Mathematics Grade 8 Learner's Book pg. 119 - Ruler - Pair of compasses - Protractor - Pencil - Plain paper	- Observation - Practical construction - Written assignments	
8	2	4.0: Geometry	4.1: Geometrical Constructions - Constructing perpendicular using set square and ruler	**At the end of the lesson, the learner should be able to:** **Knowledge:** Describe the steps for constructing perpendiculars using set square **Skills:** Construct perpendicular lines using set square and ruler **Attitudes:** Show appreciation for geometric tools	**In groups, learners are guided to:** • Draw a horizontal line • Mark point above the line • Place ruler along the line • Position set square along ruler • Slide set square until edge touches the point • Draw perpendicular line through the point	What are practical applications of perpendicular lines in construction?	- MASTER Mathematics Grade 8 Learner's Book pg. 119 - Set square - Ruler - Pencil - Drawing paper	- Observation - Practical construction - Peer review	
8	3	4.0: Geometry	4.1: Geometrical Constructions - Proportional division of a line 4.1: Geometrical Constructions - Sum of interior angles of polygons	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the method of dividing a line proportionally **Skills:** Apply the method of proportional division to divide lines into equal parts **Attitudes:** Demonstrate accuracy in geometric constructions	**In groups, learners are guided to:** • Draw line of given length • Draw auxiliary line at suitable angle • Mark equal intervals along auxiliary line using compasses • Join last point to end of original line • Draw parallel lines through other points • Verify equal divisions on original line	How can we divide a line without measuring its length?	- MASTER Mathematics Grade 8 Learner's Book pg. 119 - Ruler - Pair of compasses - Set square - Pencil - Protractor - Calculator	- Observation - Practical tasks - Written tests	
8	4	4.0: Geometry	4.1: Geometrical Constructions - Exterior angles of polygons 4.1: Geometrical Constructions - Constructing regular triangles	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define exterior angles of polygons **Skills:** Calculate sum of exterior angles and size of each exterior angle in regular polygons **Attitudes:** Appreciate the constant sum of exterior angles	**In groups, learners are guided to:** • Draw polygons and measure exterior angles • Calculate sum of exterior angles • Verify sum equals one complete revolution • Calculate exterior angle of regular polygons using formula • Complete table of polygon properties	Why is the sum of exterior angles always constant for any polygon?	- MASTER Mathematics Grade 8 Learner's Book pg. 119 - Protractor - Ruler - Calculator - Chart showing polygon properties - Pair of compasses - Pencil	- Observation - Written tests - Problem-solving tasks	
8	5	4.0: Geometry	4.1: Geometrical Constructions - Constructing regular quadrilaterals (squares)	**At the end of the lesson, the learner should be able to:** **Knowledge:** Describe properties of squares **Skills:** Construct a square using ruler and compasses **Attitudes:** Demonstrate accuracy in perpendicular construction	**In groups, learners are guided to:** • Draw line of given length • Construct perpendicular at one end using compasses • Mark point along perpendicular • Use both ends as centers to locate fourth vertex • Join points to form square • Measure angles to verify right angles at each vertex	How do we ensure all angles in a square are right angles using only compass and ruler?	- MASTER Mathematics Grade 8 Learner's Book pg. 119 - Ruler - Pair of compasses - Protractor - Plain paper	- Observation - Practical tasks - Peer assessment	
9	1	4.0: Geometry	4.1: Geometrical Constructions - Constructing regular pentagons	**At the end of the lesson, the learner should be able to:** **Knowledge:** Recall that interior angle of regular pentagon is 108° **Skills:** Construct regular pentagon using ruler and protractor **Attitudes:** Show patience in multi-step constructions	**In groups, learners are guided to:** • Draw line of given length • Measure specified interior angle at one end • Mark point along the line at given distance • Repeat process at each new vertex • Join last vertex to starting point to complete pentagon • Verify all sides and angles are equal	Why is each interior angle of a regular pentagon 108°?	- MASTER Mathematics Grade 8 Learner's Book pg. 119 - Ruler - Protractor - Pencil - Calculator	- Observation - Practical construction - Written tests	
9	2	4.0: Geometry	4.1: Geometrical Constructions - Constructing regular pentagons	**At the end of the lesson, the learner should be able to:** **Knowledge:** Recall that interior angle of regular pentagon is 108° **Skills:** Construct regular pentagon using ruler and protractor **Attitudes:** Show patience in multi-step constructions	**In groups, learners are guided to:** • Draw line of given length • Measure specified interior angle at one end • Mark point along the line at given distance • Repeat process at each new vertex • Join last vertex to starting point to complete pentagon • Verify all sides and angles are equal	Why is each interior angle of a regular pentagon 108°?	- MASTER Mathematics Grade 8 Learner's Book pg. 119 - Ruler - Protractor - Pencil - Calculator	- Observation - Practical construction - Written tests	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
9	3	4.0: Geometry	4.1: Geometrical Constructions - Constructing regular hexagons and circles	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify that interior angle of regular hexagon is 120° **Skills:** Construct regular hexagon and circles related to triangles **Attitudes:** Appreciate relationship between circles and polygons	**In groups, learners are guided to:** • Construct regular hexagon using protractor • Construct triangle and draw perpendicular bisectors • Locate circumcenter and draw circumcircle • Construct angle bisectors to find incenter and draw incircle • Compare properties of different circles	How are circles related to regular polygons and triangles?	- MASTER Mathematics Grade 8 Learner's Book pg. 119 - Ruler - Protractor - Pair of compasses - Pencil	- Observation - Practical construction - Oral questions	
9	4	4.0: Geometry	4.2: Coordinates and Graphs - Drawing labelled Cartesian plane	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define Cartesian plane and identify its components **Skills:** Draw and label a Cartesian plane with axes and origin **Attitudes:** Show understanding of coordinate system	**In groups, learners are guided to:** • Draw horizontal line and label as x-axis • Draw vertical line crossing at center and label as y-axis • Mark intersection point as origin • Number axes with positive and negative values • Place arrows at ends of axes • Discuss purpose of arrows	Why do we need two axes to locate points on a plane?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Pencil - Digital resources	- Observation - Oral questions - Written assignments	
9	5	4.0: Geometry	4.2: Coordinates and Graphs - Drawing Cartesian plane with different scales	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the concept of scale in graphs **Skills:** Draw Cartesian plane with specified scales on both axes **Attitudes:** Demonstrate accuracy in scaling	**In groups, learners are guided to:** • Draw Cartesian plane with various scales • Practice with different unit representations • Label axes correctly with chosen scale • Discuss when to use different scales • Compare graphs with different scales	How does scale affect the appearance of a graph?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Pencil - Calculator	- Observation - Practical tasks - Written tests	
10	1	4.0: Geometry	4.2: Coordinates and Graphs - Identifying points on Cartesian plane	**At the end of the lesson, the learner should be able to:** **Knowledge:** Describe how to read coordinates of points **Skills:** Read coordinates of points on Cartesian plane correctly **Attitudes:** Show precision in reading coordinates	**In groups, learners are guided to:** • Draw Cartesian plane and mark points • Draw vertical line from point to x-axis to read x-coordinate • Draw horizontal line from point to y-axis to read y-coordinate • Write coordinates with x-value first, then y-value • Practice reading multiple points in different quadrants	How do we describe the exact position of a point on a plane?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Pencil - Worksheet with points	- Observation - Oral questions - Written assignments	
10	2	4.0: Geometry	4.2: Coordinates and Graphs - Identifying points on Cartesian plane	**At the end of the lesson, the learner should be able to:** **Knowledge:** Describe how to read coordinates of points **Skills:** Read coordinates of points on Cartesian plane correctly **Attitudes:** Show precision in reading coordinates	**In groups, learners are guided to:** • Draw Cartesian plane and mark points • Draw vertical line from point to x-axis to read x-coordinate • Draw horizontal line from point to y-axis to read y-coordinate • Write coordinates with x-value first, then y-value • Practice reading multiple points in different quadrants	How do we describe the exact position of a point on a plane?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Pencil - Worksheet with points	- Observation - Oral questions - Written assignments	
10	3	4.0: Geometry	4.2: Coordinates and Graphs - Plotting points on Cartesian plane	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the process of plotting coordinates **Skills:** Plot given coordinates on Cartesian plane accurately **Attitudes:** Demonstrate accuracy in plotting	**In groups, learners are guided to:** • Identify x-coordinate and locate on x-axis • Check sign of y-coordinate • Draw line upward for positive y, downward for negative y • Locate y-coordinate on y-axis • Mark point where lines meet • Practice plotting points in all quadrants	How do we use coordinates to mark exact positions?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Pencil - List of coordinates	- Observation - Practical tasks - Peer assessment	
10	4	4.0: Geometry	4.2: Coordinates and Graphs - Reading coordinates from graphs	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify coordinates of plotted points on graphs **Skills:** Read coordinates from all four quadrants correctly **Attitudes:** Show accuracy in coordinate reading	**In groups, learners are guided to:** • Examine graph with plotted points • Write down coordinates of labeled points • Identify points on x-axis and y-axis • Match given coordinates to labeled points on graph • Practice with various coordinate positions	What special coordinates do points on the axes have?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper with plotted points - Ruler - Pencil - Practice worksheets	- Observation - Written tests - Oral questions	
10	5	4.0: Geometry	4.2: Coordinates and Graphs - Generating table of values from linear equations	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the process of generating tables from equations **Skills:** Generate table of values from given linear equations **Attitudes:** Show systematic approach to problem-solving	**In groups, learners are guided to:** • Choose suitable x values • Draw table with selected x values • Substitute each x value into equation to find y • Complete table with corresponding y values • Practice with equations in different forms	How do we find ordered pairs that satisfy a linear equation?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Calculator - Pencil	- Observation - Written assignments - Problem-solving tasks	
11	1	4.0: Geometry	4.2: Coordinates and Graphs - Completing tables for linear equations	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify given values in equation tables **Skills:** Complete given tables using equations accurately **Attitudes:** Demonstrate algebraic skills in context	**In groups, learners are guided to:** • Complete tables for equations in various forms • Substitute given values to find missing values • Generate complete tables for different equations • Practice with whole numbers and fractions • Verify completed tables	How do different forms of equations affect table generation?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Calculator - Pencil - Exercise book - Practice worksheets	- Observation - Written tests - Oral questions	
11	2	4.0: Geometry	4.2: Coordinates and Graphs - Completing tables for linear equations	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify given values in equation tables **Skills:** Complete given tables using equations accurately **Attitudes:** Demonstrate algebraic skills in context	**In groups, learners are guided to:** • Complete tables for equations in various forms • Substitute given values to find missing values • Generate complete tables for different equations • Practice with whole numbers and fractions • Verify completed tables	How do different forms of equations affect table generation?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Calculator - Pencil - Exercise book - Practice worksheets	- Observation - Written tests - Oral questions	
11	3	4.0: Geometry	4.2: Coordinates and Graphs - Determining appropriate scale for graphs	**At the end of the lesson, the learner should be able to:** **Knowledge:** List factors to consider when choosing scales **Skills:** Choose suitable scales for given data ranges **Attitudes:** Show judgment in scale selection	**In groups, learners are guided to:** • Examine table with range of values • Consider graph paper size • Calculate range of values • Select scale that accommodates all values • Ensure efficient use of graph space	How do we choose a scale that makes best use of graph paper?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Calculator - Data tables	- Observation - Practical tasks - Problem-solving	
11	4	4.0: Geometry	4.2: Coordinates and Graphs - Drawing line graphs from tables	**At the end of the lesson, the learner should be able to:** **Knowledge:** Recall steps for drawing line graphs **Skills:** Draw straight lines through plotted points using appropriate scale **Attitudes:** Show accuracy in graphing	**In groups, learners are guided to:** • Generate table of values using given equation • Choose suitable scale • Plot coordinates on Cartesian plane • Join plotted points using ruler • Draw line graphs for various equations • Verify line passes through all points	Why do linear equations produce straight line graphs?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Pencil - Calculator	- Observation - Practical construction - Peer assessment	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
11	5	4.0: Geometry	4.2: Coordinates and Graphs - Drawing graphs for various linear equations	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify equations representing horizontal and vertical lines **Skills:** Draw graphs for equations in different forms **Attitudes:** Demonstrate graphing skills	**In groups, learners are guided to:** • Draw graphs for equations in various forms • Draw horizontal and vertical lines • Compare slopes of different lines • Identify parallel and perpendicular lines • Practice graphing multiple equations	What do certain equations represent graphically?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Pencil - Set of equations	- Observation - Written tests - Practical tasks	
12	1	4.0: Geometry	4.2: Coordinates and Graphs - Introduction to simultaneous equations graphically	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define simultaneous equations **Skills:** Identify point of intersection of two lines as solution **Attitudes:** Show interest in graphical methods	**In groups, learners are guided to:** • Solve simultaneous equations algebraically • Draw graphs of both equations on same axes • Identify where lines intersect • Read coordinates of intersection point • Compare graphical solution to algebraic solution	How can graphs help us solve two equations together?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Calculator - Number cards	- Observation - Oral questions - Written assignments	
12	2	4.0: Geometry	4.2: Coordinates and Graphs - Solving simultaneous linear equations graphically	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the graphical method for solving simultaneous equations **Skills:** Solve simultaneous equations using graphs accurately **Attitudes:** Demonstrate systematic approach	**In groups, learners are guided to:** • Generate tables for both equations • Choose appropriate scale for both equations • Plot both lines on same Cartesian plane • Identify point of intersection accurately • Write solution as ordered pair • Verify solution satisfies both equations	Why must the solution satisfy both equations?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Calculator - Pencil	- Observation - Problem-solving - Written tests	
12	3	4.0: Geometry	4.2: Coordinates and Graphs - Solving simultaneous linear equations graphically	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the graphical method for solving simultaneous equations **Skills:** Solve simultaneous equations using graphs accurately **Attitudes:** Demonstrate systematic approach	**In groups, learners are guided to:** • Generate tables for both equations • Choose appropriate scale for both equations • Plot both lines on same Cartesian plane • Identify point of intersection accurately • Write solution as ordered pair • Verify solution satisfies both equations	Why must the solution satisfy both equations?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Calculator - Pencil	- Observation - Problem-solving - Written tests	
12	4	4.0: Geometry	4.2: Coordinates and Graphs - Practice solving simultaneous equations with different forms	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify equations with decimal and fractional coefficients **Skills:** Solve various forms of simultaneous equations graphically **Attitudes:** Show proficiency in graphical methods	**In groups, learners are guided to:** • Solve equations with integer coefficients • Work with decimal coefficients • Handle equations with fractions • Practice with different forms • Compare solutions for accuracy	How do decimal coefficients affect the graphing process?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Scientific calculator - Pencil	- Observation - Written tests - Problem-solving tasks	
12	5	4.0: Geometry	4.2: Coordinates and Graphs - Applying simultaneous equations to real-life problems	**At the end of the lesson, the learner should be able to:** **Knowledge:** State real-life situations involving simultaneous equations **Skills:** Formulate and solve simultaneous equations from word problems graphically **Attitudes:** Appreciate practical applications of mathematics	**In groups, learners are guided to:** • Formulate equations from shopping scenarios • Set up equations from pricing problems • Solve using graphical method • Interpret solutions in context • Discuss other real-life applications	How do simultaneous equations help solve everyday problems?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Calculator - Real-life problem cards	- Observation - Problem-solving - Oral questions	

JUNIOR SCHOOL LEARNING HUB

GRADE 8 MATHEMATICS — SCHEME OF WORK — TERM 3 2027

School: _____ Teacher: _____ TSC No.: _____

CBC planning format | Lesson outcomes are organised into Knowledge, Skills and Attitudes. Learning experiences retain the activities in the supplied Grade 8 scheme.

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
2	1	4.0: Geometry	4.3: Scale Drawing - Representation of length to given scale	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define scale and its purpose **Skills:** Determine scale from given measurements **Attitudes:** Show understanding of proportion	**In groups, learners are guided to:** • Compare sizes of objects and their representations • Discuss need for scale in drawings • Measure actual dimensions • Choose appropriate scale for representations • Calculate scale from given information • Express scale in different forms	Why do we need scale when drawing large objects?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Tape measure - Pencil - Drawing paper	- Observation - Oral questions - Practical tasks	
2	2	4.0: Geometry	4.3: Scale Drawing - Converting actual length to scale length	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the formula for converting actual length to scale length **Skills:** Convert actual measurements to scale measurements accurately **Attitudes:** Demonstrate computational skills	**In groups, learners are guided to:** • Apply given scales to convert measurements • Complete tables converting actual to scale lengths • Calculate scale lengths using various scales • Work with different units • Practice systematic conversions	How do we calculate scale length from actual length?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Ruler - Conversion tables - Pencil	- Observation - Written assignments - Problem-solving	
2	3	4.0: Geometry	4.3: Scale Drawing - Converting actual length to scale length	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the formula for converting actual length to scale length **Skills:** Convert actual measurements to scale measurements accurately **Attitudes:** Demonstrate computational skills	**In groups, learners are guided to:** • Apply given scales to convert measurements • Complete tables converting actual to scale lengths • Calculate scale lengths using various scales • Work with different units • Practice systematic conversions	How do we calculate scale length from actual length?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Ruler - Conversion tables - Pencil	- Observation - Written assignments - Problem-solving	
2	4	4.0: Geometry	4.3: Scale Drawing - Converting scale length to actual length	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the process of converting scale to actual measurements **Skills:** Convert scale measurements to actual measurements accurately **Attitudes:** Show systematic calculation approach	**In groups, learners are guided to:** • Measure lengths on scale diagrams • Use given scales to find actual lengths • Calculate actual distances • Work with different unit conversions • Practice reverse calculations	How do we find real dimensions from scale drawings?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Calculator - Scale drawings - Pencil	- Observation - Written tests - Practical tasks	
2	5	4.0: Geometry	4.3: Scale Drawing - Interpreting linear scales in statement form	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define linear scale **Skills:** Interpret scale markings and express in statement form **Attitudes:** Show understanding of scale representation	**In groups, learners are guided to:** • Examine linear scales on maps and plans • Measure length of scale markings • Determine what distance each unit represents • Practice with different linear scales • Express linear scales using words	How do linear scales differ from numerical scales?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Maps with linear scales - Ruler - Pencil - Sample plans	- Observation - Oral questions - Written assignments	
3	1	4.0: Geometry	4.3: Scale Drawing - Writing linear scales in statement form	**At the end of the lesson, the learner should be able to:** **Knowledge:** Recall the format for writing scales in statement form **Skills:** Express scales in statement form clearly and accurately **Attitudes:** Demonstrate understanding of scale notation	**In groups, learners are guided to:** • Express given scales in statement form • Write statements using proper format • Practice with scales showing various divisions • Convert linear scales to statements • Discuss advantages of statement form	Why is statement form useful for describing scales?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Linear scale examples - Pencil - Drawing paper	- Observation - Written tests - Practical tasks	
3	2	4.0: Geometry	4.3: Scale Drawing - Interpreting linear scales in ratio form	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define ratio form of scales **Skills:** Convert measurements to same units and express scales as ratios **Attitudes:** Show understanding of proportional relationships	**In groups, learners are guided to:** • Convert scales ensuring same units • Express scales as ratios • Practice unit conversions before writing ratios • Work with various scales • Understand ratios have no units	What does a scale ratio tell us about a drawing?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Calculator - Conversion charts - Pencil	- Observation - Problem-solving - Oral questions	
3	3	4.0: Geometry	4.3: Scale Drawing - Interpreting linear scales in ratio form	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define ratio form of scales **Skills:** Convert measurements to same units and express scales as ratios **Attitudes:** Show understanding of proportional relationships	**In groups, learners are guided to:** • Convert scales ensuring same units • Express scales as ratios • Practice unit conversions before writing ratios • Work with various scales • Understand ratios have no units	What does a scale ratio tell us about a drawing?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Calculator - Conversion charts - Pencil	- Observation - Problem-solving - Oral questions	
3	4	4.0: Geometry	4.3: Scale Drawing - Writing linear scales in ratio form	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the requirements for writing scales in ratio form **Skills:** Write scales in ratio form correctly without units **Attitudes:** Demonstrate accuracy in conversions	**In groups, learners are guided to:** • Complete tables converting statement to ratio form • Convert scales with various measurements • Write map scales in ratio form • Calculate ratios for different scenarios • Practice systematic conversions	How do we ensure accuracy when converting to ratio form?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Conversion tables - Pencil - Practice worksheets	- Observation - Written assignments - Problem-solving	
3	5	4.0: Geometry	4.3: Scale Drawing - Converting scale from statement to ratio form	**At the end of the lesson, the learner should be able to:** **Knowledge:** List the steps for converting statement to ratio form **Skills:** Convert statement form scales to ratio form systematically **Attitudes:** Show computational proficiency	**In groups, learners are guided to:** • Convert statement scales to ratio form • Practice with different unit combinations • Apply systematic conversion process • Work with plans and maps • Verify conversions	What steps ensure correct conversion from statement to ratio?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Ruler - Unit conversion chart - Pencil	- Observation - Written tests - Practical tasks	
4	1	4.0: Geometry	4.3: Scale Drawing - Converting scale from ratio to statement form	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the process of converting ratio to statement form **Skills:** Convert ratio form scales to statement form using appropriate units **Attitudes:** Demonstrate understanding of both forms	**In groups, learners are guided to:** • Convert ratio scales to statement form • Determine appropriate units for actual measurements • Express scales clearly in words • Practice with various ratio scales • Choose suitable units for statements	How do we choose appropriate units in statement form?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Atlas - Calculator - Ruler - Pencil	- Observation - Problem-solving - Oral questions	

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4	2	4.0: Geometry	4.3: Scale Drawing - Converting scale from ratio to statement form	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the process of converting ratio to statement form **Skills:** Convert ratio form scales to statement form using appropriate units **Attitudes:** Demonstrate understanding of both forms	**In groups, learners are guided to:** • Convert ratio scales to statement form • Determine appropriate units for actual measurements • Express scales clearly in words • Practice with various ratio scales • Choose suitable units for statements	How do we choose appropriate units in statement form?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Atlas - Calculator - Ruler - Pencil	- Observation - Problem-solving - Oral questions	
4	3	4.0: Geometry	4.3: Scale Drawing - Making scale drawings with calculations	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify dimensions needed for scale drawings **Skills:** Calculate scale lengths and make accurate scale drawings **Attitudes:** Show precision in measurements and drawing	**In groups, learners are guided to:** • Calculate scale lengths before drawing • Make accurate scale drawings of various shapes • Apply appropriate scales • Measure and verify dimensions • Calculate areas from scale drawings	Why must we calculate scale lengths before drawing?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Pencil - Calculator - Drawing paper	- Observation - Practical construction - Written tests	
4	4	4.0: Geometry	4.3: Scale Drawing - Scale drawings with distance calculations	**At the end of the lesson, the learner should be able to:** **Knowledge:** Recall how to measure distances on drawings **Skills:** Make scale drawings involving multiple distances and calculate actual distances **Attitudes:** Show systematic approach to problem-solving	**In groups, learners are guided to:** • Make scale drawings involving multiple points • Use suitable scales for given distances • Measure lengths on scale drawings • Calculate actual distances from drawings • Apply geometric principles where needed • Verify measurements	How do scale drawings help solve distance problems?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Pair of compasses - Calculator - Graph paper	- Observation - Practical tasks - Problem-solving	
4	5	4.0: Geometry	4.3: Scale Drawing - Using maps and demonstrating scale	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify scales on actual maps **Skills:** Read scales from maps and measure distances accurately **Attitudes:** Appreciate real-world applications of scale	**In groups, learners are guided to:** • Examine maps in atlas • Identify and read map scales • Measure distances between locations • Calculate actual distances using scale • Compare different maps with different scales • Discuss map features	How does scale choice affect what we can show on a map?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Atlas - Maps - Ruler - Calculator - Digital resources	- Observation - Practical measurement - Oral questions	
5	1	4.0: Geometry	4.3: Scale Drawing - Application problems with scale	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify given information in scale problems **Skills:** Solve complex problems involving scale, area, volume, time and speed **Attitudes:** Show advanced problem-solving skills	**In groups, learners are guided to:** • Solve problems involving height and scale • Find scales used in given scenarios • Calculate areas from scale diagrams • Determine time and speed using map scales • Work with various measurement scenarios • Apply multiple concepts together	How do professionals use scale in their work?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Ruler - Problem cards - Reference materials	- Observation - Problem-solving - Written tests	
5	2	4.0: Geometry	4.3: Scale Drawing - Application problems with scale	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify given information in scale problems **Skills:** Solve complex problems involving scale, area, volume, time and speed **Attitudes:** Show advanced problem-solving skills	**In groups, learners are guided to:** • Solve problems involving height and scale • Find scales used in given scenarios • Calculate areas from scale diagrams • Determine time and speed using map scales • Work with various measurement scenarios • Apply multiple concepts together	How do professionals use scale in their work?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Ruler - Problem cards - Reference materials	- Observation - Problem-solving - Written tests	
5	3	4.0: Geometry	4.3: Scale Drawing - Using ICT for scale and maps	**At the end of the lesson, the learner should be able to:** **Knowledge:** Describe how digital maps use scale **Skills:** Use digital devices to display maps and demonstrate zoom functions **Attitudes:** Show digital literacy in geography context	**In groups, learners are guided to:** • Access digital maps on devices • Use zoom function to change scale • Observe how scale changes with zoom level • Measure distances on digital maps • Compare scale indicators on digital and paper maps • Discuss advantages of digital tools	How does zooming affect the scale of a digital map?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Digital devices (tablets/computers) - Internet access - Digital mapping software - Projector	- Observation - Practical demonstration - Oral questions	
5	4	4.0: Geometry	4.4: Common Solids - Identifying common solids from environment	**At the end of the lesson, the learner should be able to:** **Knowledge:** Name common solids: cubes, cuboids, cylinders, pyramids and cones **Skills:** Classify solids by their properties **Attitudes:** Show awareness of geometric shapes in environment	**In groups, learners are guided to:** • Collect objects from environment • Group objects by shape categories • Identify properties of each solid type • Discuss examples in daily life • Create display of classified solids	Where do we see these solids in our daily lives?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Collection of solid objects - Models of solids - Pictures of buildings - Digital images	- Observation - Practical classification - Oral questions	
5	5	4.0: Geometry	4.4: Common Solids - Properties of solids (faces, edges, vertices)	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define faces, edges and vertices **Skills:** Identify and count faces, edges and vertices of given solids **Attitudes:** Show understanding of 3D properties	**In groups, learners are guided to:** • Examine labeled solids • Name all faces of solids • Identify all edges • Locate all vertices • Practice with different solids • Record properties systematically	How do faces, edges and vertices define a solid?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Models of solids - Ruler - Labels - Worksheet	- Observation - Written assignments - Practical identification	
6	1	4.0: Geometry	4.4: Common Solids - Sketching nets of cubes	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define the term "net" of a solid **Skills:** Sketch nets of closed and open cubes **Attitudes:** Demonstrate spatial visualization	**In groups, learners are guided to:** • Label cube vertices • Cut cube along specified edges • Lay out faces on flat surface • Sketch net showing all faces for closed cube • Sketch net showing appropriate faces for open cube • Identify different possible net arrangements	How does a 3D cube transform into a 2D net?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cubes - Scissors/razor blade - Ruler - Pencil - Plain paper	- Observation - Practical construction - Peer assessment	
6	2	4.0: Geometry	4.4: Common Solids - Sketching nets of cubes	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define the term "net" of a solid **Skills:** Sketch nets of closed and open cubes **Attitudes:** Demonstrate spatial visualization	**In groups, learners are guided to:** • Label cube vertices • Cut cube along specified edges • Lay out faces on flat surface • Sketch net showing all faces for closed cube • Sketch net showing appropriate faces for open cube • Identify different possible net arrangements	How does a 3D cube transform into a 2D net?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cubes - Scissors/razor blade - Ruler - Pencil - Plain paper	- Observation - Practical construction - Peer assessment	

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6	3	4.0: Geometry	4.4: Common Solids - Sketching nets of cuboids	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify faces of cuboids **Skills:** Sketch nets of closed and open cuboids **Attitudes:** Show accuracy in net construction	**In groups, learners are guided to:** • Label cuboid vertices • Cut along specified edges • Spread faces on flat surface • Sketch net with all faces for closed cuboid • Sketch net with appropriate faces for open cuboid • Identify pairs of equal faces	How does the net of a cuboid differ from that of a cube?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cuboids - Scissors/razor blade - Ruler - Pencil - Grid paper	- Observation - Practical tasks - Written tests	
6	4	4.0: Geometry	4.4: Common Solids - Sketching nets of cylinders	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify components of cylinder nets **Skills:** Sketch nets of closed and open cylinders **Attitudes:** Understand curved surface as rectangle	**In groups, learners are guided to:** • Cut cylinder to remove bases • Cut along height to open curved surface • Observe curved surface forms rectangle • Note rectangle dimensions relate to circumference • Sketch nets for closed and open cylinders • Label components	Why does the curved surface become a rectangle?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cylinders - Scissors/razor blade - Ruler - Pair of compasses - Pencil	- Observation - Practical construction - Oral questions	
6	5	4.0: Geometry	4.4: Common Solids - Sketching nets of pyramids	**At the end of the lesson, the learner should be able to:** **Knowledge:** Describe components of pyramid nets **Skills:** Sketch nets of pyramids with different bases **Attitudes:** Show precision in drawing nets	**In groups, learners are guided to:** • Label pyramid vertices • Cut along slant edges • Lay faces on flat surface • Sketch net showing base and triangular faces • Ensure triangular faces connect to base edges • Practice with different base dimensions	How many triangular faces does a square-based pyramid have?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model pyramids - Scissors/razor blade - Ruler - Pencil - Drawing paper	- Observation - Practical tasks - Peer review	
7	1	4.0: Geometry	4.4: Common Solids - Sketching nets of cones	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify components of cone nets **Skills:** Sketch nets of cones showing sector shape **Attitudes:** Appreciate relationship between arc and circumference	**In groups, learners are guided to:** • Cut base from cone • Cut curved surface along slant height • Observe curved surface forms sector • Note relationship between arc length and base circumference • Sketch net showing circle and sector • Label components	Why does the cone's curved surface form a sector?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cones - Scissors/razor blade - Protractor - Pair of compasses - Pencil	- Observation - Practical construction - Written assignments	
7	2	4.0: Geometry	4.4: Common Solids - Sketching nets of cones	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify components of cone nets **Skills:** Sketch nets of cones showing sector shape **Attitudes:** Appreciate relationship between arc and circumference	**In groups, learners are guided to:** • Cut base from cone • Cut curved surface along slant height • Observe curved surface forms sector • Note relationship between arc length and base circumference • Sketch net showing circle and sector • Label components	Why does the cone's curved surface form a sector?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cones - Scissors/razor blade - Protractor - Pair of compasses - Pencil	- Observation - Practical construction - Written assignments	
7	3	4.0: Geometry	4.4: Common Solids - Matching solids to nets and vice versa	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify solids from their nets **Skills:** Match given solids to correct nets **Attitudes:** Demonstrate spatial reasoning	**In groups, learners are guided to:** • Match various solids to their nets • Identify which solid each net will form • Draw solid that corresponds to given net • Practice visualizing 3D from 2D • Sketch nets for solids with various dimensions	How can we visualize the solid from looking at its net?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Various nets - Model solids - Ruler - Pencil - Matching cards	- Observation - Practical matching - Problem-solving	
7	4	4.0: Geometry	4.4: Common Solids - Surface area of cubes from nets	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the formula for surface area of cube **Skills:** Calculate total surface area of cube from its net **Attitudes:** Show systematic calculation approach	**In groups, learners are guided to:** • Measure sides of cube • Sketch net of cube • Calculate area of one face • Multiply by number of faces • Practice with cubes of different dimensions • Verify by drawing net and calculating	How does knowing one side help find total surface area?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cubes - Ruler - Calculator - Pencil - Net templates	- Observation - Written tests - Problem-solving	
7	5	4.0: Geometry	4.4: Common Solids - Surface area of cuboids from nets	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the formula for surface area of cuboid **Skills:** Calculate total surface area of cuboid from nets **Attitudes:** Show organized calculation method	**In groups, learners are guided to:** • Draw net of cuboid with given dimensions • Calculate areas of different faces • Identify pairs of equal faces • Add all areas to find total • Practice with various dimensions • Verify calculations	Why do we calculate surface area in pairs for cuboids?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cuboids - Ruler - Calculator - Grid paper - Pencil	- Observation - Written assignments - Practical calculations	
8	1	4.0: Geometry	4.4: Common Solids - Surface area of cylinders from nets	**At the end of the lesson, the learner should be able to:** **Knowledge:** State components of cylinder surface area **Skills:** Calculate total surface area of cylinder from nets **Attitudes:** Demonstrate formula application	**In groups, learners are guided to:** • Identify net components • Calculate area of circular faces • Find rectangle dimensions using circumference • Calculate rectangular area • Add areas for total surface area • Practice with different dimensions	How is the circumference used in finding surface area?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cylinders - Ruler - Calculator - Pair of compasses - Formula chart	- Observation - Problem-solving - Written tests	
8	2	4.0: Geometry	4.4: Common Solids - Surface area of cylinders from nets	**At the end of the lesson, the learner should be able to:** **Knowledge:** State components of cylinder surface area **Skills:** Calculate total surface area of cylinder from nets **Attitudes:** Demonstrate formula application	**In groups, learners are guided to:** • Identify net components • Calculate area of circular faces • Find rectangle dimensions using circumference • Calculate rectangular area • Add areas for total surface area • Practice with different dimensions	How is the circumference used in finding surface area?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cylinders - Ruler - Calculator - Pair of compasses - Formula chart	- Observation - Problem-solving - Written tests	
8	3	4.0: Geometry	4.4: Common Solids - Surface area of pyramids from nets	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify components of pyramid surface area **Skills:** Calculate total surface area of pyramid from nets **Attitudes:** Show systematic approach to complex calculations	**In groups, learners are guided to:** • Draw net showing base and triangular faces • Calculate base area • Calculate area of each triangular face • Add base area to sum of triangular areas • Practice with different dimensions • Verify calculations	How do we find the slant height if not given?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model pyramids - Ruler - Calculator - Pencil - Net templates	- Observation - Written assignments - Problem-solving	

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8	4	4.0: Geometry	4.4: Common Solids - Surface area of cones and distance on surfaces	**At the end of the lesson, the learner should be able to:** **Knowledge:** State formula for surface area of cone **Skills:** Calculate surface area of cone from net and determine shortest distances on solid surfaces **Attitudes:** Show advanced spatial reasoning	**In groups, learners are guided to:** • Identify cone net components • Calculate circular base area • Calculate sector area using given angle • Find total surface area • Open cuboid into net to find paths between points • Measure distances along net surface	How does opening a solid help find distances on its surface?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cones and cuboids - Protractor - Calculator - String - Scissors	- Observation - Problem-solving - Practical tasks	
8	5	4.0: Geometry	4.4: Common Solids - Surface area of cones and distance on surfaces	**At the end of the lesson, the learner should be able to:** **Knowledge:** State formula for surface area of cone **Skills:** Calculate surface area of cone from net and determine shortest distances on solid surfaces **Attitudes:** Show advanced spatial reasoning	**In groups, learners are guided to:** • Identify cone net components • Calculate circular base area • Calculate sector area using given angle • Find total surface area • Open cuboid into net to find paths between points • Measure distances along net surface	How does opening a solid help find distances on its surface?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cones and cuboids - Protractor - Calculator - String - Scissors	- Observation - Problem-solving - Practical tasks	
9	1	4.0: Geometry	4.4: Common Solids - Making models of hollow solids (cubes and cuboids)	**At the end of the lesson, the learner should be able to:** **Knowledge:** List steps for making hollow models **Skills:** Construct hollow cube and cuboid models from nets **Attitudes:** Show craftsmanship in model making	**In groups, learners are guided to:** • Draw nets accurately on manila paper • Include flaps for joining faces • Cut out nets carefully • Fold along marked lines • Paste flaps to form hollow solids • Display completed models	Why do we need flaps when making hollow models?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Manila paper - Ruler - Pencil - Scissors - Glue/paste - Colored markers	- Observation - Practical construction - Peer assessment	
9	2	4.0: Geometry	4.4: Common Solids - Making models of cylinders, cones and pyramids	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain steps for making different hollow models **Skills:** Construct hollow cylinder, cone and pyramid models **Attitudes:** Show precision and craftsmanship	**In groups, learners are guided to:** • Draw cylinder net with calculated dimensions • Construct cone net with appropriate sector angle • Draw pyramid net with correct measurements • Cut and fold to form solids • Paste edges to complete models • Display and compare models	How do we ensure the cylinder's rectangle matches the circle?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Manila paper - Pair of compasses - Protractor - Ruler - Scissors - Glue	- Observation - Practical construction - Written reflection	
9	3	4.0: Geometry	4.4: Common Solids - Using IT devices and drawing technology	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify technology tools for learning about solids **Skills:** Use technology to explore and draw solids and nets **Attitudes:** Appreciate technology in mathematics learning	**In groups, learners are guided to:** • Watch educational videos about solids • Use software to draw 3D shapes • Explore rotating solids digitally • Practice drawing nets using technology • Use apps to visualize net folding • Share digital creations	How does technology enhance our understanding of 3D shapes?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Computers/tablets - Internet access - GeoGebra software - Projector - 3D modeling apps	- Observation - Digital portfolio - Oral presentation - Peer evaluation	
9	4	5.0: Data Handling and Probability	5.1: Data Presentation and Interpretation - Collecting data and drawing bar graphs	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define bar graph and identify its components **Skills:** Collect data from own experiences and draw bar graphs with suitable scale **Attitudes:** Appreciate the use of graphs in presenting data	**In groups, learners are guided to:** • Collect data from class members on given characteristics • Fill data in tables • Choose suitable scale for collected data • Draw bar graphs to represent collected data • Compare graphs with other groups • Discuss components of bar graphs	How can we represent collected data visually?	- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Ruler - Graph paper - Pencil - Data collection sheets	- Observation - Practical tasks - Oral questions	
9	5	5.0: Data Handling and Probability	5.1: Data Presentation and Interpretation - Drawing bar graphs with suitable scale 5.1: Data Presentation and Interpretation - Interpreting bar graphs	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the steps for drawing bar graphs **Skills:** Draw bar graphs with appropriate scales for different data sets **Attitudes:** Show accuracy in graph construction	**In groups, learners are guided to:** • Choose uniform width for bars • Select uniform gaps between bars • Choose suitable scale for vertical axis • Calculate heights of bars according to scale • Draw bars accurately • Label axes properly • Practice with various data sets	How do we choose an appropriate scale for a bar graph?	- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Graph paper - Ruler - Pencil - Calculator - Data tables - Sample bar graphs - Question sheets	- Observation - Practical construction - Written assignments	
10	1	5.0: Data Handling and Probability	5.1: Data Presentation and Interpretation - Drawing line graphs	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define line graph and state its uses **Skills:** Draw line graphs from given data **Attitudes:** Appreciate line graphs for showing trends	**In groups, learners are guided to:** • Choose suitable scale for x-axis • Choose suitable scale for y-axis • Plot points from table of values • Join plotted points using straight lines • Label axes appropriately • Practice drawing line graphs for different data sets	When is it appropriate to use a line graph instead of a bar graph?	- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Graph paper - Ruler - Pencil - Calculator - Data tables	- Observation - Practical construction - Peer assessment	
10	2	5.0: Data Handling and Probability	5.1: Data Presentation and Interpretation - Interpreting line graphs 5.1: Data Presentation and Interpretation - Identifying mode of discrete data	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain how to read values from line graphs **Skills:** Interpret line graphs to extract information **Attitudes:** Show analytical skills in reading trends	**In groups, learners are guided to:** • Read values at specific points on graph • Identify highest and lowest values • Determine trends from line graphs • Calculate totals from graph data • Answer questions based on line graphs • Discuss patterns observed	How do line graphs help us see changes over time?	- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Sample line graphs - Ruler - Pencil - Question sheets - Number cards - Exercise books - Data sets	- Observation - Written tests - Problem-solving	
10	3	5.0: Data Handling and Probability	5.1: Data Presentation and Interpretation - Calculating mean of discrete data 5.1: Data Presentation and Interpretation - Working out averages from different sets	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the formula for calculating mean **Skills:** Calculate the mean of discrete data sets accurately **Attitudes:** Show systematic approach in calculations	**In groups, learners are guided to:** • Note down values from group members • Add all values in data set • Count number of values • Divide sum by number of values • Calculate mean for various data sets • Verify calculations • Practice with different contexts	How does the mean represent the average of a data set?	- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Calculator - Pencil - Exercise books - Data sets - Problem cards	- Observation - Written tests - Problem-solving	
10	4	5.0: Data Handling and Probability	5.1: Data Presentation and Interpretation - Determining median of discrete data	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define median and explain the process of finding it **Skills:** Determine the median of discrete data for odd and even sets **Attitudes:** Show systematic approach in finding median	**In groups, learners are guided to:** • Arrange data in ascending or descending order • Identify middle value for odd sets • Calculate median for even sets by averaging two middle values • Practice finding median for various data sets • Compare median with mode and mean • Discuss applications	Why must data be arranged in order before finding the median?	- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Number cards - Pencil - Exercise books - Calculator	- Observation - Oral questions - Written tests	

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10	5	5.0: Data Handling and Probability	5.1: Data Presentation and Interpretation - Using IT for data presentation and calculations	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify IT tools for creating graphs **Skills:** Use technology to create bar graphs and line graphs and calculate mean, mode and median **Attitudes:** Appreciate technology in data handling	**In groups, learners are guided to:** • Use spreadsheet software to enter data • Create bar graphs using software • Create line graphs using software • Use formulas to calculate mean • Use functions to find mode and median • Compare manual and digital methods • Present findings digitally	How does technology make data presentation and analysis easier?	- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Computers/tablets - Spreadsheet software - Internet access - Projector - Data sets	- Observation - Digital portfolio - Practical demonstration - Peer evaluation	
11	1	5.0: Data Handling and Probability	5.2: Probability - Identifying events involving chance in real life	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define chance and probability **Skills:** Identify events involving chance in daily life **Attitudes:** Show awareness of probability in real situations	**In groups, learners are guided to:** • Discuss possibilities in various scenarios • Identify chance events in sports • Recognize chance in weather predictions • Discuss chance in games • List daily events involving chance • Share observations with class	What is chance and where do we encounter it in daily life?	- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Pictures of chance events - Pencil - Chart paper - Real-life scenario cards	- Observation - Oral questions - Class discussion	
11	2	5.0: Data Handling and Probability	5.2: Probability - Identifying events involving chance in real life	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define chance and probability **Skills:** Identify events involving chance in daily life **Attitudes:** Show awareness of probability in real situations	**In groups, learners are guided to:** • Discuss possibilities in various scenarios • Identify chance events in sports • Recognize chance in weather predictions • Discuss chance in games • List daily events involving chance • Share observations with class	What is chance and where do we encounter it in daily life?	- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Pictures of chance events - Pencil - Chart paper - Real-life scenario cards	- Observation - Oral questions - Class discussion	
11	3	5.0: Data Handling and Probability	5.2: Probability - Discussing likely and unlikely events	**At the end of the lesson, the learner should be able to:** **Knowledge:** List the likelihood scale terms: impossible, unlikely, equally likely, likely, certain **Skills:** Classify events as impossible, unlikely, equally likely, likely or certain **Attitudes:** Show critical thinking in analyzing probability	**In groups, learners are guided to:** • Examine likelihood scale • Discuss meaning of each term • Classify statements using likelihood terms • Identify impossible events • Identify certain events • Distinguish between likely and unlikely • Practice with various statements	How do we describe the likelihood of different events happening?	- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Likelihood scale chart - Event cards - Pencil - Exercise books	- Observation - Oral questions - Written assignments	
11	4	5.0: Data Handling and Probability	5.2: Probability - Performing chance experiments	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define chance experiment **Skills:** Perform chance experiments such as flipping coins, tossing dice, and drawing objects **Attitudes:** Show interest in hands-on probability activities	**In groups, learners are guided to:** • Obtain coins and flip them • Toss dice and record outcomes • Draw colored balls or beads from bags • Use spinners and record results • Record outcomes from experiments • Compare results with other groups • Discuss patterns observed	What are the possible outcomes when we perform chance experiments?	- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Coins - Dice - Colored balls/beads - Bags - Spinners - Recording sheets	- Observation - Practical tasks - Oral questions	
11	5	5.0: Data Handling and Probability	5.2: Probability - Writing experimental probability outcomes	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the concept of experimental probability **Skills:** Write all possible outcomes from chance experiments **Attitudes:** Demonstrate systematic recording of outcomes	**In groups, learners are guided to:** • List possible outcomes from coin toss • Write outcomes from die roll • Determine outcomes from spinners • List outcomes from drawing objects • Form combinations of outcomes • Record outcomes systematically • Share findings with class	How do we list all possible outcomes from an experiment?	- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Coins - Dice - Number cards - Pencil - Exercise books	- Observation - Written tests - Problem-solving	
12	1	5.0: Data Handling and Probability	5.2: Probability - Expressing probability outcomes as fractions	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the formula for probability as a fraction **Skills:** Express probability outcomes as fractions accurately **Attitudes:** Show understanding of favorable outcomes	**In groups, learners are guided to:** • Identify total possible outcomes • Identify favorable outcomes • Express probability as fraction of favorable to total outcomes • Simplify probability fractions • Calculate probabilities from various scenarios • Solve word problems involving probability • Verify answers	How do we express the chance of an event happening as a fraction?	- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Colored balls/beads - Bags - Calculator - Pencil - Exercise books	- Observation - Written assignments - Problem-solving tasks	
12	2	5.0: Data Handling and Probability	5.2: Probability - Expressing probability outcomes as fractions	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the formula for probability as a fraction **Skills:** Express probability outcomes as fractions accurately **Attitudes:** Show understanding of favorable outcomes	**In groups, learners are guided to:** • Identify total possible outcomes • Identify favorable outcomes • Express probability as fraction of favorable to total outcomes • Simplify probability fractions • Calculate probabilities from various scenarios • Solve word problems involving probability • Verify answers	How do we express the chance of an event happening as a fraction?	- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Colored balls/beads - Bags - Calculator - Pencil - Exercise books	- Observation - Written assignments - Problem-solving tasks	
12	3	5.0: Data Handling and Probability	5.2: Probability - Expressing probability as decimals and percentages	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the relationship between probability in fractions, decimals and percentages **Skills:** Convert probability from fractions to decimals and percentages **Attitudes:** Demonstrate proficiency in probability conversions	**In groups, learners are guided to:** • Convert probability fractions to decimals • Convert probability fractions to percentages • Understand that probability in decimals cannot exceed 1 • Understand that probability in percentages cannot exceed 100% • Calculate complementary probabilities • Solve problems in different forms • Apply probability in real contexts	Why is probability sometimes expressed as decimals or percentages?	- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Calculator - Pencil - Exercise books - Conversion charts	- Observation - Written tests - Problem-solving	
12	4	5.0: Data Handling and Probability	5.2: Probability - Using IT to play probability games	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify digital tools for probability activities **Skills:** Use technology to play games involving probability and simulate experiments **Attitudes:** Appreciate technology in learning probability	**In groups, learners are guided to:** • Access online probability games • Use software to simulate coin flips • Use apps to simulate dice rolls • Play digital probability games • Record results from digital experiments • Compare manual and digital experiments • Discuss advantages of using technology	How does technology help us understand probability better?	- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Computers/tablets - Internet access - Probability apps/software - Projector - Recording sheets	- Observation - Digital portfolio - Practical demonstration - Oral presentation	

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12	5	5.0: Data Handling and Probability	5.2: Probability - Using IT to play probability games	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the key concept in the sub-strand. **Skills:** Apply the concept in mathematical tasks and real-life situations. **Attitudes:** Appreciate the usefulness of mathematics in everyday life.	**In groups, learners are guided to:** • Discuss the key ideas and relate them to familiar situations. • Demonstrate the relevant mathematical procedure using suitable learning resources. • Practise the skills and present findings to peers.				